

Diabetes Prediction using Machine Learning

UST — 100 Next Gen AI Engineer Cohort 3 | Assignment 2 • End-to-end classification pipeline with Scikit-learn

1. Dataset Overview

The Pima Indians Diabetes dataset contains **768 patient records** across **9 columns** (8 medical attributes + the target). The target variable Outcome is binary: 0 = No Diabetes, 1 = Diabetes. Class distribution: **500** patients without diabetes vs **268** with diabetes — a moderate class imbalance (~65% / 35%).

2. Data Cleaning & Preprocessing

- **Hidden missing values:** Glucose, BloodPressure, SkinThickness, Insulin and BMI cannot be 0 in a real patient, so 0s in these columns were treated as missing: Glucose (5), BloodPressure (35), SkinThickness (227), Insulin (374), BMI (11). These were imputed using the **median** of each column.
- **Train-test split:** 614 training rows / 154 testing rows (80/20 split, stratified to preserve the class ratio).
- **Feature scaling:** features were standardized (zero mean, unit variance) for Logistic Regression, which is sensitive to feature scale. Decision Tree and Random Forest were trained on the raw (unscaled) features, since tree-based models split on thresholds and do not require scaling.
- No categorical encoding was required — all input features are already numeric.

3. Model Building & Training

Three classification algorithms were implemented and compared: **Logistic Regression** (linear baseline, max_iter=1000), **Decision Tree** (max_depth=5), and **Random Forest** (n_estimators=100, max_depth=6). Tree depth was limited on purpose to reduce overfitting risk.

Model	Train Accuracy	Test Accuracy	Precision	Recall	F1-Score
Logistic Regression	79.64%	70.78%	60.00%	50.00%	0.545
Decision Tree	81.43%	75.97%	63.93%	72.22%	0.678
Random Forest	88.44%	74.03%	66.67%	51.85%	0.583

Overfitting/underfitting observation: Random Forest shows the largest gap between train accuracy (88.4%) and test accuracy (74.0%), indicating mild overfitting. The Decision Tree's train/test gap is smaller, and Logistic Regression — being a simple linear model — shows the most consistent (but comparatively lower) performance across train and test.

4. Model Evaluation (Test Set)

The **Decision Tree** performed best overall on the test set (highest F1-Score = 0.678). Its confusion matrix breaks down as: **True Negatives = 78**, **False Positives = 22**, **False Negatives = 15**, **True Positives = 39**.

This means the model correctly identified 39 of 54 actual diabetic patients (Recall = 72.2%), and 22 healthy patients were incorrectly flagged as diabetic (false alarms).

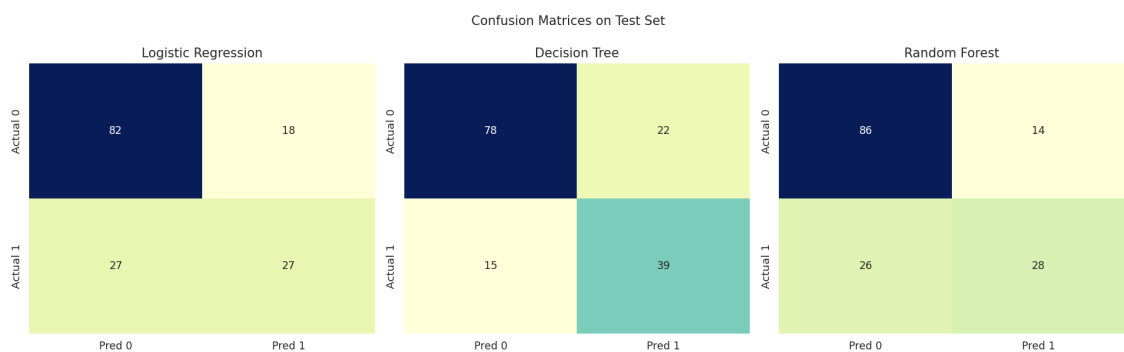


Fig 1. Confusion matrices for all three models on the held-out test set.

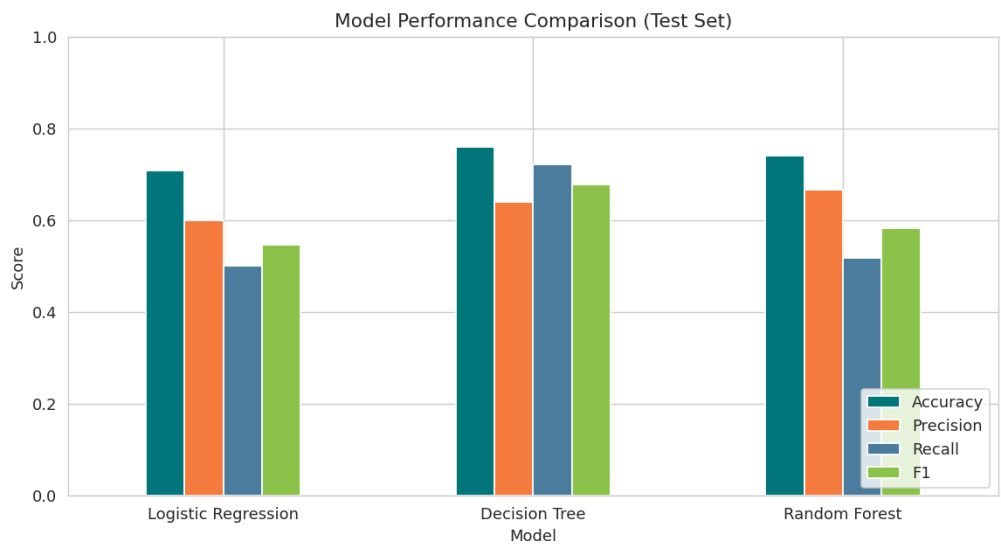


Fig 2. Accuracy, Precision, Recall and F1-Score compared across the three models on the test set.

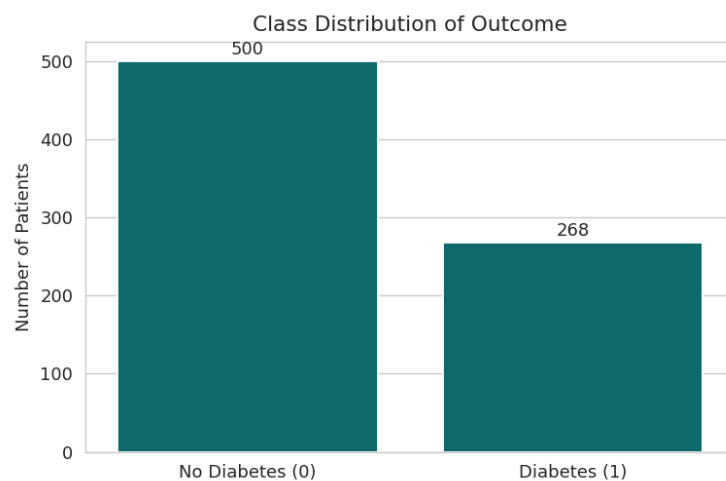


Fig 3. Class distribution of the target variable — a moderate imbalance toward "No Diabetes".

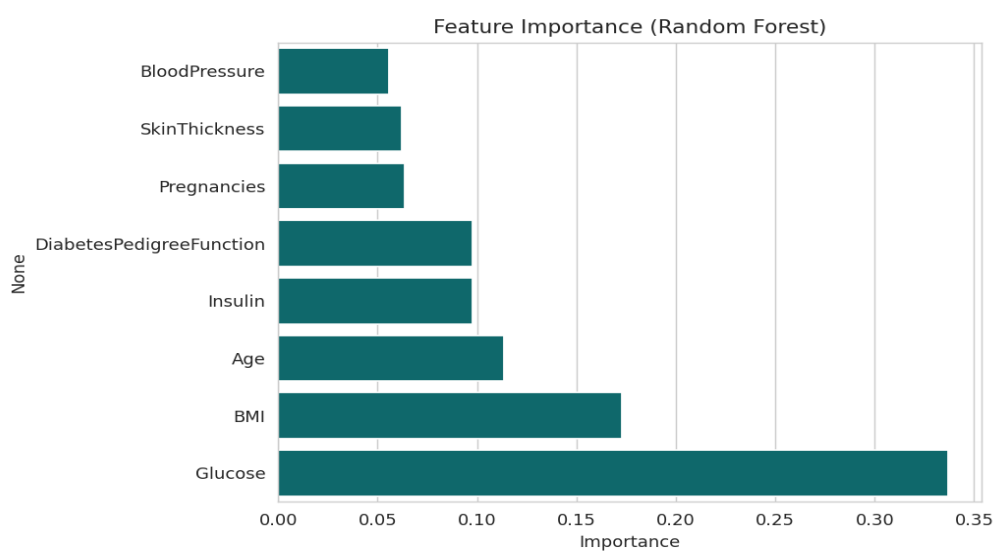


Fig 4. Random Forest feature importance. Top predictors: Glucose (33.7%), BMI (17.3%), Age (11.4%).

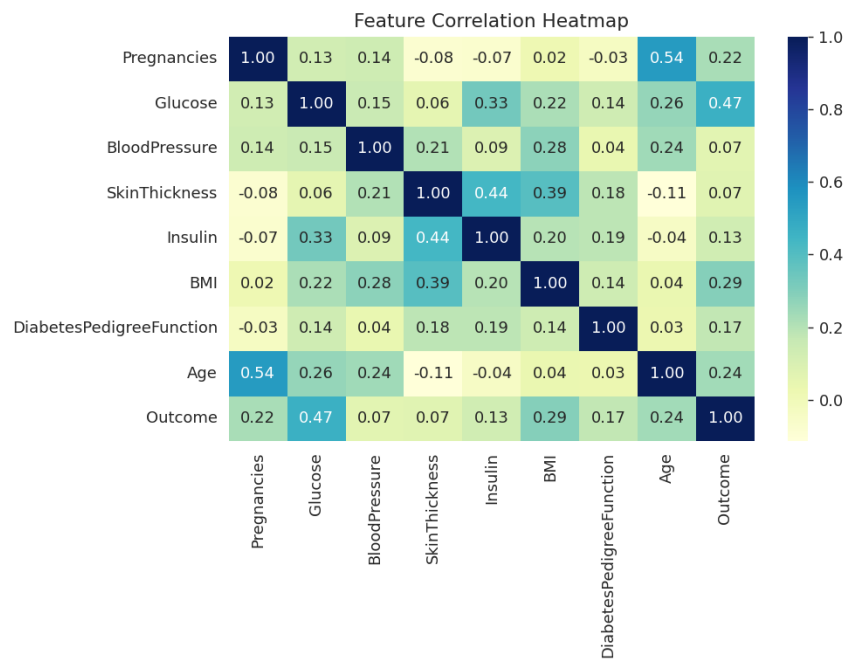


Fig 5. Correlation heatmap of all features and the target. Glucose has the strongest correlation with Outcome.

5. Model Performance Analysis & Conclusion

Most important metric: For a diabetes screening model, **Recall** (sensitivity) matters most. A False Negative means telling a diabetic patient they are healthy, delaying treatment — a more costly error than a False Positive (an extra confirmatory test). We therefore prioritize Recall, balanced against Precision via the F1-Score, over raw Accuracy.

Is the model performing well? The Decision Tree is the best of the three (Accuracy 76.0%, Recall 72.2%, F1 0.678) — reasonable but not excellent for a real medical screening tool. Roughly 1 in 4 diabetic patients in the test set would still be missed (False Negatives). Random Forest had the highest training accuracy but generalized slightly worse here and had lower Recall; Logistic Regression was the most consistent across train/test but had the lowest Recall of the three.

Limitations: the dataset is relatively small (768 records) and moderately imbalanced; several features (Insulin, SkinThickness) required heavy imputation (over 29% and 48% of values missing respectively), introducing noise; and the data reflects only female Pima Indian patients, limiting generalization to other populations.

Recommendations to improve performance:

1. **Handle class imbalance** — e.g. SMOTE oversampling or `class_weight='balanced'`, to improve Recall on the minority (diabetic) class.
2. **Hyperparameter tuning** — GridSearchCV/RandomizedSearchCV over tree depth, number of estimators, and minimum samples per leaf, especially to reduce Random Forest overfitting.
3. Additional options: feature engineering (e.g. BMI/Glucose bins), feature selection to drop low-importance features (BloodPressure, SkinThickness), and trying gradient boosting algorithms (e.g. XGBoost).